



Hands-on Exercise No. 2
DigiSkills 3.0 Batch-03
AI Using Python

Total Marks: 10
Due Date: 14/05/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
 - It is submitted after due date.
 - It is not in the required format (.doc or .docx)
 - It does not open, or file is corrupt.
 - It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this exercise, you shall be able to:

- Load and explore datasets using **Pandas DataFrame**, including viewing data, checking structure, and generating summary statistics.
- Perform **data cleaning and preprocessing** by handling missing values, removing duplicates, selecting relevant features, and encoding categorical data.
- Prepare a structured dataset suitable for **machine learning tasks**, building a foundation for model training and evaluation in real-world scenarios.

Problem Statement

In this hands-on-exercise, you are required to perform **data preparation and preprocessing** using Python and Pandas on the given dataset (**HousePricePrediction.csv**).

You will load the dataset into a DataFrame, explore its structure, and understand the data through basic analysis. After that, you will clean the dataset by handling missing values and removing duplicates, followed by selecting relevant features and preparing the data for machine learning by converting categorical variables into numerical form.

The objective is to prepare a clean and structured dataset that can be used for further analysis and model building in the next hands-on-exercise.

You can use any programming environment such as **Jupyter Notebook, Google Colab, VS Code, or any other Python editor** to write and execute your code.

After completing each task, take **screenshots of your code and its output**, and paste them into a **single MS Word (.doc/.docx) file** for submission.

Tasks:

Download the **HousePricePrediction** file from LMS by the following link:

<https://lms.digiskills.pk/Courses/AIP301/Downloads/HousePricePrediction.csv>

1. Dataset Loading: **(Marks 2)**
 - Load the dataset into a Pandas DataFrame
 - Display the first 10 rows

2. Data Exploration: **(Marks 2)**
 - Display the shape of the dataset
 - Check data types of all columns
 - Generate summary statistics of numerical features

3. Data Cleaning: **(Marks 2)**
 - Identify missing values
 - Handle missing values appropriately
 - Check and remove duplicate records if any

4. Feature Selection: **(Marks 2)**
 - Identify relevant features for analysis
 - Remove unnecessary columns such as identifiers (e.g., Id)

5. Data Preprocessing: **(Marks 2)**
 - Convert categorical variables into numerical form (encoding)
 - Preparing the dataset for machine learning and display it's preview in the output

NOTE: Save the data file and code file at your end for Hands-on Exercise 3.

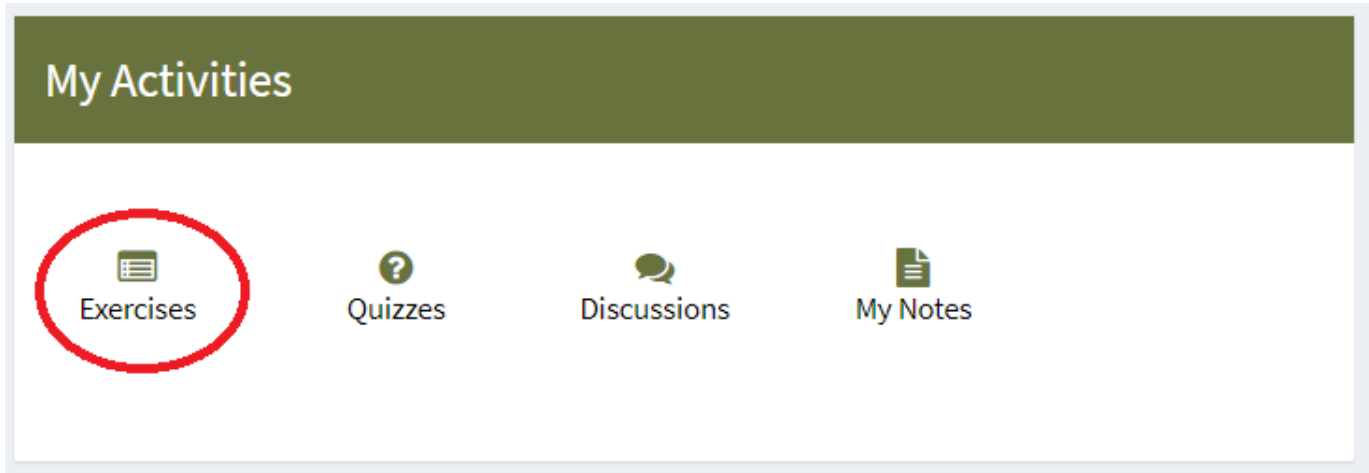
In this solution file, no need to send/attach code file; only provide screenshots of the above tasks.

BEST OF LUCK ☺

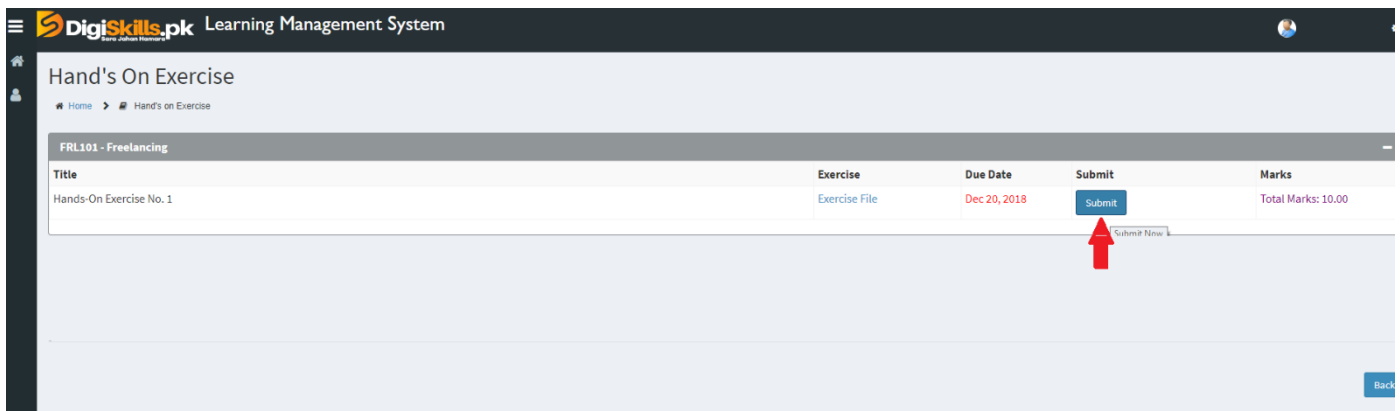
How to submit solution file on LMS?

Please perform the following steps for submitting your solution using LMS:

- 1) Login to the LMS
- 2) Click on the **Exercises** button within the **My Activities** section



- 3) Click on the submit button to upload your Solution.



- 4) Keep in mind to upload your Solution in .doc or .docx format